RANK-ONE HORROCKS OVER A LOCAL RING BY DETERMINANTS

CLAUDE (ANTHROPIC, OPUS 5.5)

Abstract. Let A be a local ring and L an ideal of $A[X]$ containing a monic polynomial, such that the inclusion $L \subseteq A[X]$ has an A-linear retraction. We give a short proof that L is generated by a monic polynomial: $A[X]/L$ is free over A, the characteristic polynomial G of multiplication by X lies in L, and a surjection between free modules of the same finite rank is an isomorphism, so $L = (G)$. Each step is constructive. The companion note [2] proves the same theorem by division, without freeness or determinants.

1. Statement

All rings are commutative. A is local: if $a + b$ is a unit, then a or b is a unit. We do not require $0 \neq 1$. For a monic $F \in A[X]$ of degree d, every p has a unique decomposition $p = Fq + r$ with $\deg r < d$, and q and r are A-linear in p.

Theorem 1. *Let L be an ideal of $A[X]$ containing a monic polynomial F, and let $\rho : A[X] \to L$ be A-linear with $\rho(p) = p$ for $p \in L$. Then L is generated by a monic polynomial, and this polynomial is unique.*

Uniqueness holds over any ring: two monic polynomials dividing each other are equal [2, Lemma 8]. The proof of existence has three steps: $A[X]/L$ is free of some rank e (Section 2); the characteristic polynomial G of multiplication by X on it is monic of degree e and lies in L (Section 3); and every element of L of degree $< e$ is 0, so G divides L (Section 4).

2. Freeness

Let V be an A-module and $N \subseteq V$ a submodule; write $p \equiv p'$ for $p - p' \in N$. A *presentation* of V/N by m generators consists of $g_1, \dots, g_m \in V$ and A-linear forms $\kappa_1, \dots, \kappa_m$ on V with

$$p \equiv \textstyle\sum_t \kappa_t(p)\, g_t \quad \text{for all } p \in V, \qquad \kappa_t(p) = 0 \quad \text{for } p \in N.$$

It is a *basis* if moreover $\kappa_i(g_j) = \delta_{ij}$. Then $\sum_t c_t g_t \in N$ forces $c = 0$, by applying κ.

The forms κ say that V/N is projective: they define $V/N \to A^m$, and $p \mapsto \sum_t \kappa_t(p) g_t$ splits it. The next lemma is the local freeness lemma (see [5]), in this language.

Lemma 2. *If A is local, then V/N has a basis.*

Proof. Induction on m. For $m = 0$ the empty family is a basis. Let $m > 0$ and $c = \kappa_m(g_m)$. Since $c + (1 - c) = 1$, c or $1 - c$ is a unit.

If $1 - c$ is a unit, then $g_m \equiv c g_m + \sum_{t<m} \kappa_t(g_m) g_t$ gives $g_m \equiv \sum_{t<m} (1-c)^{-1} \kappa_t(g_m) g_t$. Substituting this for g_m, the generators $g_1, \dots, g_{m-1}$ with the forms $\kappa'_t(p) = \kappa_t(p) + \kappa_m(p)(1-c)^{-1}\kappa_t(g_m)$ present V/N, and induction applies.

If c is a unit, put $k(p) = c^{-1}\kappa_m(p)$ and $p' = p - k(p) g_m$, so that $k(g_m) = 1$ and $\kappa_m(p') = 0$. Let $N' = \{p \mid p' \in N\}$, the submodule $N + A g_m$. The generators $g_1, \dots, g_{m-1}$ with the forms $p \mapsto \kappa_t(p')$ present V/N': for $p \in V$, the presentation of V/N applied to p' gives $p' \equiv \sum_{t<m} \kappa_t(p') g_t =: y$, hence $k(y) = k(p') = 0$ and $(p - y)' = p' - y \in N$. By induction

V/N' has a basis $b_1, \dots, b_{e'}$ with forms λ'. Then $g_m, b'_1, \dots, b'_{e'}$ with the forms k, λ' is a basis of V/N. Both forms vanish on $N \subseteq N'$. For $p \in V$, with $z = p - \sum_j \lambda'_j(p) b_j \in N'$,

$$p - k(p) g_m - \textstyle\sum_j \lambda'_j(p)\, b'_j = z' \in N,$$

and $k(b'_j) = 0$, $\lambda'(g_m) = 0$ (since $g_m \in N'$) and $\lambda'(b'_j) = \lambda'(b_j)$ give the duality relations. □

Locality gives a mere disjunction, so the basis merely exists. This suffices here, since the monic generator is unique.

Application. Take $V = A[X]$ and $N = L$. Let $F \in L$ be monic of degree d and r the remainder modulo F. The monomials $1, X, \dots, X^{d-1}$ with $\kappa(p) =$ the coefficients of $r(p - \rho(p))$ present $A[X]/L$: indeed $p - r(p - \rho(p)) = \rho(p) + Fq(p - \rho(p)) \in L$, and $p \in L$ gives $p - \rho(p) = 0$. So $A[X]/L$ has a basis $b_1, \dots, b_e$ with forms $\lambda_1, \dots, \lambda_e$.

3. The characteristic polynomial

We use two facts about determinants over a commutative ring B: the adjugate satisfies $\operatorname{adj}(M)M = \det(M)\, I$, and $\det(MM') = \det M \det M'$. The first gives the determinant trick.

Lemma 3 (determinant trick, [1, Prop. 2.4])**.** *Let W be a B-module, $M \in M_n(B)$ and $w_1, \dots, w_n \in W$ with $\sum_j M_{ij} w_j = 0$ for all i. Then $\det(M)\, w_k = 0$ for all k.*

Proof. $\det(M)\, w_k = \sum_j (\operatorname{adj}(M)M)_{kj} w_j = \sum_i \operatorname{adj}(M)_{ki} \sum_j M_{ij} w_j = 0.$ □

Lemma 4. *For $a \in M_n(A)$, $\det(XI - a) \in A[X]$ is monic of degree n.*

Proof. Say that p has degree $\le k$ and leading coefficient c if $p - cX^k$ has degree $< k$. This relation is preserved by sums, and degree $\le k$ times degree ≤ 1 gives degree $\le k + 1$ with the product of the leading coefficients. The entries of $XI - a$ have degree ≤ 1 with leading coefficients the entries of I; expanding the determinant as a sum of products of n entries, $\det(XI - a)$ has degree $\le n$ and leading coefficient $\det I = 1$. □

Make A^e an $A[X]$-module through the basis: $p \cdot v = \lambda(p \sum_j v_j b_j)$. This is the module $A[X]/L$ in coordinates; the module laws hold because $\equiv$ is compatible with multiplication. Let a be the matrix of multiplication by X acting on rows, $Xb_i \equiv \sum_j a_{ij} b_j$, that is $a_{ij} = \lambda_j(Xb_i)$, and let $e_1, \dots, e_e$ be the standard basis of A^e. Then

$$\textstyle\sum_j (X\delta_{ij} - a_{ij}) \cdot e_j = \lambda\big(Xb_i - \sum_j a_{ij} b_j\big) = 0,$$

so by Lemma 3, $G = \det(XI - a)$ kills every e_k, hence all of A^e. Since $1 \equiv \sum_k \lambda_k(1) b_k$,

$$\lambda(G) = \lambda(G \cdot 1) = G \cdot \lambda(1) = 0,$$

so $G \equiv \sum_k \lambda_k(G) b_k = 0$: $G \in L$. By Lemma 4, G is monic of degree e. This is the Cayley–Hamilton theorem for multiplication by X on $A[X]/L$.

4. A surjection between free modules

Let $S \in M_e(A)$ be the matrix of $c \mapsto \lambda(\sum_k c_k X^k)$, from the coefficients of polynomials of degree $< e$ to A^e. It has a right inverse: let T be the matrix whose j-th column is the coefficients of the remainder $\bar{b}_j$ of b_j modulo G. Since $G \in L$, $\bar{b}_j \equiv b_j$, so ST has j-th column $\lambda(\bar{b}_j) = \lambda(b_j) = e_j$: $ST = I$. Hence $\det S \det T = 1$, $\det S$ is a unit, and $\operatorname{adj}(S)S = \det(S)\, I$ shows that S is injective (this is Lemma 3 for $W = A$).

Now let $p \in L$ and $p = Gq + r$ with $\deg r < e$. Then $r = p - Gq \in L$, so $\lambda(r) = 0$, that is, S kills the coefficients of r, and $r = 0$. So G divides every element of L, which proves Theorem 1.

Remark 5. The rank e is the degree of the monic generator, and locality is used only in Lemma 2. The rank-one Horrocks theorem follows as in [2, §4]: an invertible ideal containing a monic polynomial has a retraction ρ, built from division by a monic polynomial.

5. Comparison

The argument combines three classical facts: finitely generated projective modules over a local ring are free [5], the determinant trick [1, Prop. 2.4], and the fact that a surjection between free modules of the same finite rank is an isomorphism. We do not know a reference for this combination applied to Horrocks' theorem. The three proofs of Horrocks' theorem in [4, Ch. IV] proceed differently.

Swan [4, §IV.3]. From $P_f \cong F_f$ with $F = F_0[t]$ and f monic, Swan builds a Towber presentation: with D the elements of F of degree $< \deg f$, the module $M = D \cap P \cong P/fF$ is projective over R by Schanuel's lemma, t induces α on M, and $(t-\alpha)m = f\beta(m)$ defines $\beta : M \to F_0$. The key step is that β is nonzero modulo the maximal ideal $\mathfrak{m}$, proved over the residue field from the characteristic sequence of $t - \alpha$; then an element x with $\beta(x) \notin \mathfrak{m}F_0$ is split off, using that projective modules over the local ring are free, and induction applies. Swan's proof shares the freeness step with ours, but has no determinant step.

Roberts [4, §IV.4]. An axiomatic argument (Thm. IV.4.1): bases of P modulo $\mathfrak{m}$ and of $S^{-1}P$ are matched using a factorisation of $GL_n(S^{-1}A)$, and Nakayama's lemma, applied to modules finitely generated over R because f is monic, gives a basis of P. No determinants are used.

Nashier–Nichols [4, §IV.5]. In rank one (Prop. IV.5.2, for invertible ideals I), a monic generator of the image of I in $k[t]$ is lifted to a monic $f \in I$ (Lemma IV.5.1), the equality $I \cap \mathfrak{m}[t] = \mathfrak{m}I$ is shown using I^{-1}, and Nakayama's lemma applied to I/Af, which is finitely generated over R, gives $I = Af$. Neither freeness of $A[t]/I$ nor determinants are used.

Lombardi–Quitté [5, §XVI.4]. The local Horrocks theorem (Thm. XVI.4.3, for a residually discrete local ring) follows Nashier–Nichols for the induction on the rank, with Nakayama's lemma and the comaximality of the monic polynomials with $1 + \mathfrak{m}[X]$. Rank one is treated by the lemma of Murthy and Pedrini (Lemma XVI.4.6): if $P_S \cong Q_S$, Schanuel's lemma and the freeness of $A[X]/(f)$ over A for f monic make P and Q stably isomorphic, and two stably isomorphic modules of rank one are isomorphic through the top exterior power, $P \cong \bigwedge^{k+1}(P \oplus A^k)$ (Fact X.5.6). Determinants thus appear there too, but through stable isomorphism and exterior powers, not through freeness of $A[X]/L$ and a characteristic polynomial.

6. Formalisation

The proof is checked in `sag/horrocks` of `https://github.com/coquand/unit`, with the same statement as the division proof (`DivisorStep`, `Retract.Induction.generator`): `DeterminantHorrocks`, `Proof.generator`. Lemma 2 is `LocalFree`, for any A-module and submodule. The determinant comes from the exterior algebra of [3] (`sag/exterior`): `Determinant` ($\det(MM') = \det M \det M'$), `Adjugate` ($\mathrm{adj}(M)M = \det(M)\, I$), `ModuleTrick` (Lemma 3), `DeterminantTrick` (the case $W = B$), and `CharPoly` (Lemma 4, from a general lemma on relations preserved by the products defining the determinant).

References

[1] M. F. Atiyah and I. G. Macdonald, *Introduction to Commutative Algebra*, Addison–Wesley, 1969.

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[3] T. Coquand and C. Quitté, Constructive finite free resolutions, *Manuscripta Math.* 137 (2012), 331–345.
[4] T. Y. Lam, *Serre's Problem on Projective Modules*, Springer, 2006.
[5] H. Lombardi and C. Quitté, *Commutative Algebra: Constructive Methods*, Springer, 2015.